

RESEARCH INTERESTS

Database systems, storage systems, distributed systems

- Modern system architectures: time-series databases, disaggregated databases
- Core database techniques in transaction processing: indexing, concurrency control
- HW-SW co-design: RDMA, SmartNIC, CXL memory, byte-addressable storage

EDUCATION

University of Wisconsin-Madison

Ph.D. in Computer Science

- Advisor: Prof. Xiangyao Yu

Madison, WI
Jan. 2021 – Present

Sungkyunkwan University

M.S. in Electrical and Computer Engineering

- Thesis: Leveraging Byte-Addressability of Persistent Memory for B+-tree
- Advisor: Prof. Beomseok Nam

South Korea
Sep. 2018 – Aug. 2020

B.S. in Computer Science and Engineering

B.B.A. in Business Administration

Mar. 2012 – Aug. 2018
(incl. Korean military service)

WORK EXPERIENCE

University of Wisconsin-Madison

Research Assistant, Database Group

- Designed and implemented B^{link} -hash, a high-performance in-memory index that adapts to workload changes from heavy writes to intensive reads
- Developed a two-sided RDMA transaction processing system that minimizes network traffic and guarantees starvation freedom in disaggregated memory databases
- Designed a lightweight disk-based index that minimizes footprint and provides fast reads

Madison, WI
May 2021 – Present

Microsoft Research

Research Intern, Data Systems Group

- Supervisor: Dr. Badrish Chandramouli

Redmond, WA
Summer 2026

Microsoft

Research Intern, Gray Systems Lab (GSL)

- Supervisors: Dr. Rathijit Sen, and Dr. Matteo Interlandi
- Built a transaction processing system for disaggregated databases
- Designed a low-latency concurrency control mechanism for high-performance networks (*patent pending*)

Redmond, WA
Summer 2025

Intel

Software Engineering Intern, Data Platforms Future Technology Team

- Evaluated cost-performance analysis of storage devices in data-center system environments.
- Performed end-to-end infrastructure optimizations for database platforms

Hillsboro, OR
Summer 2022

Sungkyunkwan University

Researcher, Convergence Research Institute

- Built the prototype of *BallonStasher*, an RDMA-based disaggregated memory that improves resource utilization and system performance via memory sharing

South Korea
Sep. 2020 – Dec. 2020

Research Assistant, Data-Intensive Computing Lab

- Developed B^3 -tree, a write-optimized B+-tree for persistent memory that minimizes write traffic in maintaining data ordering by employing binary search tree logical representation

Sep. 2018 – Aug. 2020

PUBLICATIONS

- [1] *Disaggregated Memory for File-backed Pages*, Daegyul Han, Jaeyoon Nam, **Hokeun Cha**, Changdae Kim, Kwangwon Koh, Taehoon Kim, Sang-Hoon Kim, Beomseok Nam, ACM Transactions on Storage (**TOS**), Nov. 2025

	[2] <i>Towards Accelerating Data Intensive Application's Shuffle Process Using SmartNICs</i>, Jiaxin Lin, Tao Ji, Xiangpeng Hao, Hokeun Cha, Yanfang Le, Xiangyao Yu, Aditya Akella, Proceedings of ACM on Measurement and Analysis of Computing Systems (SIGMETRICS), Jun. 2023 [3] <i>B^{link}-hash: An Adaptive Hybrid Index for In-Memory Time-Series Databases</i>, Hokeun Cha, Xiangpeng Hao, Tianzheng Wang, Huanchen Zhang, Aditya Akella, Xiangyao Yu, Proceedings of VLDB Endowment, Feb. 2023 [4] <i>Pivotal B+ tree for Byte-Addressable Persistent Memory</i>, Jonghyeon Yoo, Hokeun Cha, Wonbae Kim, Wook-Hee Kim, Sung-Soon Park, Beomseok Nam, IEEE Access, Apr. 2022 [5] <i>B³-tree: Byte-Addressable Binary B-Tree for Persistent Memory</i>, Hokeun Cha, Moohyeon Nam, Kibeom Jin, Jiwon Seo, Beomseok Nam, ACM Transactions on Storage (TOS), Jul. 2020 [6] <i>Write-Optimized Dynamic Hashing for Persistent Memory</i>, Moohyeon Nam, Hokeun Cha, Young-ri Choi, Sam H. Noh, Beomseok Nam, Proceedings of 17th USENIX Conference on File and Storage Technologies (FAST), Feb. 2019	
WORK-IN-PROGRESS	[1] Revisiting Two-sided RDMA for Disaggregated Memory Databases (<i>under review</i>) [2] Block B-tree: A Lightweight Disk-Based Index for Time-Series Databases	
TALKS	<i>B^{link}-hash: An Adaptive Hybrid Index for In-Memory Time-Series Databases</i> • Database Group Seminar, Cornell University, Ithaca, NY (<i>virtual</i>) Feb. 2024 • VLDB Conference, Vancouver, Canada, Feb. 2023 • Database Affiliates Workshop, UW-Madison, Madison, WI Oct. 2022 <i>Write-Optimized Dynamic Hashing for Persistent Memory</i> • Korea Software Congress (KSC), Pyeongchang, South Korea Dec. 2019	
TEACHING EXPERIENCE	University of Wisconsin-Madison Madison, WI Teaching Assistant • CS564: Database Management Systems Spring 2021 Sungkyunkwan University South Korea Teaching Assistant • SWE3021: Multi-core Computing Fall 2019 • SWE3006: Programming Languages Spring 2019	
HONORS	Computer Science Departmental Scholarship Spring 2021 – Fall 2021 University of Wisconsin-Madison, Madison, WI Merit-based Scholarship for Graduate Students Fall 2018 – Spring 2020 Sungkyunkwan University, Seoul, South Korea Merit-based Scholarship for Undergraduate Students Spring 2013 Sungkyunkwan University, Seoul, South Korea	
SERVICE	Reviewing Activities Journal of Data Science and Engineering 2026 Journal of Cluster Computing 2024 Journal of Supercomputing 2023, 2024	